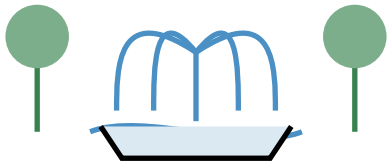
Chapter 11

Public Goods and Common Resources



When Charging Each User Is Difficult

Let's imagine a fireworks display that many people can enjoy at once. Once it is visible, preventing nonpaying neighbors from watching may be difficult.

Now let's compare this with fish in an open-access ocean. Reserving them for paying users is also difficult, but a fish caught by one person is unavailable to others.

Both goods are hard to exclude people from using, but only the fish are used up by consumption. This difference changes the incentive problem.

Excludability and Rivalry

To make this distinction precise, let's separate the ability to restrict access from the effect of one person's use on others.

Definition. A good is **excludable** if people can be prevented from using it.

A seller can withhold a meal from someone who does not pay.

Definition. A good is **rival in consumption** if one person's use reduces another person's ability to use it.

Eating a meal leaves less food for others.

These are properties of access and use. They do not tell us whether a government or a private firm supplies the good.

Four Kinds of Goods

Combining excludability and rivalry gives four categories.

	Rival	Not rival
Excludable	Private goods	Club goods
	A meal	*Subscription streaming*
Not excludable	Common resources	Public goods
	Open-access fish	*National defense*

For streaming, we assume capacity is sufficient so another viewer does not reduce the service available to others.

The Classification Depends on Conditions

A good's category can change with its access and use conditions. For example, let's consider a quiet, untolled road. It is neither excludable nor rival in use: adding one driver does not delay anyone else.

When the road becomes congested, an additional driver slows others down. Use is now rival.

An enforceable toll makes access excludable. Thus, the same road can fall into different categories depending on congestion and access rules.

Discussion

How would we classify each road: as a private good, club good, public good, or common resource? Let's explain both excludability and rivalry.

1. An uncongested road with no toll.
2. An uncongested road with an enforceable toll.
3. A congested road with no toll.
4. A congested road with an enforceable toll.

Answer

The road examples illustrate how congestion and tolls determine the category.

1. The uncongested untolled road is a public good. Access is not restricted, and another driver does not reduce others' use.

2. The uncongested toll road is a club good. Access can be restricted, but use is not rival.

3. The congested untolled road is a common resource. Access is open, and another driver delays others.

4. The congested toll road is a private good in this classification. Access is restricted, and use is rival.

Example: Shared Facilities at NTU

Let's consider two hypothetical campus facilities. Their classification depends on access rules and crowding, not on the fact that a university provides them.

- A study room requires a valid student card. Access is excludable. When many desks are empty, another user need not reduce anyone else's use, so the room can function as a club good.
- An open bicycle rack has no enforceable access restriction. When it is full, one person's bicycle occupies space someone else could use, making it a common resource under these conditions.

What Changes When the Facility Fills Up?

When the study room fills up, another student can use a desk only if someone else gives it up. Use becomes rival, while card access remains excludable. The room now fits the private-good category.

Restricting access to the bicycle rack would make it excludable, but would not create more spaces. Access control and physical capacity are different margins.

For discussion, does making a crowded facility free turn it into a public good? No. A zero price does not eliminate rivalry or determine whether nonusers can be excluded.

Public Goods and Free Riding

Definition. A **public good** is a good that is neither excludable nor rival in consumption.

Nonexcludability makes voluntary financing difficult.

Definition. A **free rider** is a person who receives a benefit without paying for it.

If nonpayers cannot be excluded, each person may prefer to let others pay.

As a result, voluntary payments can fall short of the cost even when total willingness to pay exceeds that cost. The private market can underprovide the good.

Example: A Neighborhood Fountain

Let's suppose 200 residents can enjoy a park fountain without crowding, and nonpayers cannot be excluded. Each values it at NT$3,000. Construction and maintenance cost NT$240,000 for the same period.

$$\text{Total benefit} = 200 \times 3{,}000 = \text{NT\$600,000}.$$

Building the fountain adds $600,000 - 240,000 =$ NT$360,000 to total surplus, so provision is efficient.

- Yet if voluntary contributions total only NT$90,000, the project remains unfunded. Each resident may hope others will pay because enjoyment does not depend on contributing.

Thus, the fountain is worth building, yet voluntary contributions may not finance it. This raises a separate question: how can we cover its cost?

Financing the Fountain

A compulsory payment can cover the fountain's cost. Dividing it equally among residents gives:

$$\frac{\text{NT\$240,000}}{200} = \text{NT\$1,200 per resident.}$$

Each resident receives a benefit of NT\$3,000 and pays NT\$1,200, gaining NT\$1,800 in this example.

This financing result depends on two assumptions:

- Residents receive equal benefits. With unequal benefits, an efficient project need not make every resident better off under an equal tax.
- Payments can be collected at no additional cost. Otherwise, collection uses resources and reduces the project's net benefit.

Adding Benefits for the Same Public Good

Because a public good is nonrival, the same additional unit can benefit several people. Its marginal social benefit is therefore the sum of their willingness to pay for that unit.

Let's suppose Mei, Jun, and Lin live on a lane with a shared streetlight. They value its first hour of lighting after sunset at NT$30, NT$20, and NT$10, respectively. That same hour benefits all three, so its marginal social benefit is NT$60.

If that hour costs NT$45, providing it adds NT$15 of surplus. We add *values at the same quantity*, rather than adding quantities at the same price.

How Long Should the Streetlight Stay On?

Suppose only these three neighbors benefit. Their hypothetical marginal benefits in NT$ for successive hours after sunset are:

Hour	Mei	Jun	Lin	Total
First	30	20	10	60
Second	20	15	5	40
Third	10	5	0	15

Now suppose each hour costs NT$30. The first two hours each add surplus, whereas the third costs more than it benefits them. The efficient quantity is **two hours**, with total surplus $60 + 40 - 2 \times 30 = \text{NT\$40}$.

How could they fund the NT$60 cost? Mei, Jun, and Lin could pay NT$30, NT$20, and NT$10, respectively. Each pays less than their benefit from two hours. But they must agree on and collect payments: nonpayers still enjoy the light, so free riding can prevent voluntary funding.

Public Goods and Research Incentives

The same financing problem arises when benefits are shared widely and nonpayers cannot be excluded.

- **National defense** protects residents together, and excluding particular nonpayers is difficult.
- **Basic research** creates knowledge that, once openly available, can be used by another person without being used up.

Knowledge can be costly to produce even when sharing it costs almost nothing. If researchers cannot capture its benefits, private funding may be too low; public research funding can help.

An enforceable patent instead makes use of an invention excludable for a time, raising the inventor's reward while restricting access.

Poverty Reduction and Shared Benefits

A shared benefit does not make every input to its provision a public good. For example, food given to one household is a private good. The same meal cannot also be eaten by another household.

However, people may value living in a society with less poverty. That shared benefit can extend to people who do not donate.

This creates a possible free-rider problem in charitable giving. It does not mean that every good supplied through an antipoverty program is itself a public good.

Cost–Benefit Analysis

Definition. Cost–benefit analysis is a method of evaluating a project by comparing its total benefits and opportunity costs to society.

For a public good, market purchases may not reveal willingness to pay. Survey respondents may overstate benefits if someone else pays, or understate them to avoid a contribution.

A sound comparison includes maintenance, time, environmental effects, and the value of resources used. It compares feasible alternatives over the same period. A project improves welfare when its benefits exceed its costs relative to the alternative.

Discussion: Flood Protection in Taipei

Let's suppose a hypothetical drainage improvement protects 100 Taipei households from typhoon flooding. Nonpayers cannot be excluded, and protecting one home does not reduce protection for the others.

For the same period, each of 20 households values protection at NT$10,000. Each of the other 80 values it at NT$1,000. Building and maintaining the project costs NT$200,000. There are no effects outside this group.

1. Does the project pass a cost–benefit test?

2. Would every household support an equal compulsory payment? Why might voluntary contributions still fall short?

Answer: Benefits and the Financing Rule

The flood-protection service is a public good under the stated conditions. Its total benefit is

$$20 \times 10{,}000 + 80 \times 1{,}000 = \text{NT\$}280{,}000.$$

Net benefit is NT\$80,000, so the project passes the cost–benefit test.

An equal payment is NT\$2,000 per household. The 80 households with benefits of NT\$1,000 would lose NT\$1,000 each under that financing rule.

The lesson is that efficiency does not imply unanimous support for every tax allocation. Voluntary financing also faces free riding because nonpayers remain protected.

Example: Valuing a Safer Intersection

Let's suppose a hypothetical traffic improvement costs NT$600,000 and reduces each of 10,000 residents' lifetime fatality risk by 1/100,000. Each is willing to pay NT$100 for this reduction in their own risk.

- Total benefit is NT$1,000,000 (10,000 $\times$ 100). With no other effects, net benefit is NT$400,000.
- The **value of a statistical life** is willingness to pay for risk reductions per expected death prevented. Here expected deaths prevented are 10,000/100,000 = 0.1, giving 1,000,000/0.1 = NT$10 million.

This measure values many small risk reductions. It is not a price at which a particular person agrees to die or a measure of their earnings.

Common Resources and Overuse

Definition. A **common resource** is a good that is rival in consumption and nonexcludable under the prevailing access rules.

Unlike a public good, one person's use therefore leaves less for others.

- In an open-access fishery, a catch benefits the fisher but leaves fewer fish for others and may reduce future reproduction.
- On shared grazing land, an extra sheep benefits its owner but leaves less grass for other households' flocks.

Each user receives the private return while bearing only part of the social cost. The **Tragedy of the Commons** is the resulting tendency to overuse a common resource.

Example: An Extra Fishing Trip

Let's suppose an additional fishing trip brings NT$5,000 of expected revenue and costs NT$3,000 in fuel, time, and equipment use.

Other fishers lose NT$3,000 in current and future net returns because of this trip. These hypothetical values include all relevant effects.

The fisher gains 5,000 − 3,000 = NT$2,000. Social cost also includes the NT$3,000 loss to others, so the trip reduces total surplus by NT$1,000. A NT$3,000 fee would make the fisher face this social cost and forgo the trip.

Managing a Common Resource

Preventing this overuse requires changing incentives or restricting use.

- A fee can make users account for the harm caused by another unit of use.
- A quota or tradable permit system can limit total use.
- Defined ownership or community access rules can give users a reason to preserve the resource.

Each approach needs monitoring and enforcement. Rules must cover the people whose actions affect the resource.

Example: Farms Sharing Groundwater

An **aquifer** is an underground layer of rock or sediment that stores and transmits groundwater. Farmers pump this water through wells to irrigate crops, especially when rainfall is insufficient.

Let's consider farms sharing an aquifer in southern Taiwan. In this hypothetical example, pumping is not effectively restricted. As one farm pumps, the groundwater level falls: other farms must pump from greater depths or have less water for their crops.

An extra unit of pumping gives one farm NT$500 of additional crop revenue and costs it NT$200 in energy and equipment use. Suppose the falling groundwater level also causes NT$400 of losses to other farms.

The pumping farm gains $500-200 = \text{NT\$300}$, whereas total surplus changes by $500 - 200 - 400 = -\text{NT\$100}$.

Managing the Shared Aquifer

A fee of NT$400 on the extra unit would make the pumping farm account for the losses it imposes on others. Its private cost would become NT$600, exceeding the NT$500 benefit.

Alternatively, a community could limit total pumping and allocate access rights. Metering and enforcement would be needed to make the limit effective.

For discussion, would owning the land above each well solve the problem? Separate ownership of fields does not by itself give a farmer control over neighbors' extraction from the same aquifer.

Roads, Water, and Fish

The same reasoning applies to other resources whose use harms others. On a congested road, a driver bears their own travel time but also delays other drivers. A congestion charge can target that additional delay.

When a river receives waste, one user's disposal reduces water quality for others. The river's capacity to absorb waste is scarce.

Fish that move across boundaries are difficult for a single owner to protect. Cooperation matters when conservation benefits users outside one jurisdiction.

Discussion

Let's suppose a town is considering two policies:

1. Collect contributions to fund a warning siren audible to all residents.
2. Limit catches from a lake that anyone can currently fish.

For each policy, let's identify the type of good, explain the incentive problem, and describe what the policy is trying to change.

Answer

The warning service is a public good if residents cannot be excluded and hearing it does not reduce others' ability to hear it. The problem is underfunding because people can benefit without contributing.

The open-access fish stock is a common resource. A fish caught by one person is unavailable to others, so the problem is excessive use.

The first policy finances provision. The second limits use and protects the stock.

Property Rights and Market Outcomes

Definition. Property rights are enforceable rights governing who may use a resource and under what conditions.

If enforcement makes exclusion feasible, rivalry determines the new category:

- A **common resource becomes a private good** if it remains rival. An owner who controls access to a fish stock can charge for catches and benefit from preserving fish, reducing the incentive for overuse.
- A **public good becomes a club good** if it remains nonrival. A subscription broadcast can collect fees to fund provision, but excludes nonpayers who could otherwise share the service at no additional cost.

Exclusion does not guarantee efficiency. Government or community rules can also address funding or use. Ownership alone does not determine the category: a government-owned good need not be a public good.

Takeaways

Access and rivalry determine the incentive problem.

- Excludability concerns who can be kept out; rivalry concerns whether one person's use reduces another's.
- Public goods face free riding and underprovision. For each additional unit, efficiency compares the sum of users' benefits with its social cost.
- Common resources face overuse because users impose costs on others.
- Taxes, access rules, permits, and defined rights can improve incentives when they address the underlying problem.